Applied Regression Interpretation

Gov 51: Data Analysis and Politics

Scott Cunningham

Harvard University

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The Big Picture

Women earn 84 cents for every dollar men earn

Where does that number come from?

And what does it actually mean?

Regression turns data into claims you can interrogate

- **Today:** One dataset, three models, building complexity
- **Skills you need:**
  - Interpret every coefficient in context
  - Compute $\hat{Y}$ by plug-in
  - Compare models side-by-side
  - Read interactions

The Current Population Survey is the gold standard for labor data

- **What:** Monthly survey of ∼60,000 U.S. households
- **Who runs it:** Bureau of Labor Statistics + Census Bureau
- **How:** Interviewers visit or call sampled households
- **Why it exists:** Official source for unemployment rate, earnings, labor force participation
- Every jobs report you see in the news comes from the CPS

Wooldridge extracted 526 workers from the 1976 CPS

- Jeff Wooldridge compiled this sample for his econometrics textbook
- 1976: pre-Equal Pay enforcement, large raw gender gaps
- Clean, real data — every number we compute today is a fact about these 526 people

Variable	Type	Description
`wage`	Continuous	Hourly wage ($/hr)
`educ`	Continuous	Years of education
`exper`	Continuous	Years of work experience
`tenure`	Continuous	Years at current job
`female`	Binary (0/1)	= 1 if woman
`married`	Binary (0/1)	= 1 if married

$n = 526$ · 274 men, 252 women

Model 1: The Raw Gap

Model 1 starts with the population regression

$$\text{wage}_i \quad = \quad \beta_0 \; + \; \beta_1 \; \cdot \; \text{female}_i \; + \; \epsilon_i$$

- wage_i — the **outcome** (*dependent variable*): hourly wage for person i
- female_i — the **predictor** (*independent variable*): equals 1 if woman, 0 if man
- β_0, β_1 — **unknown parameters** we want to learn
- ϵ_i — the **error term**: everything else that affects wages

We estimate the unknowns using OLS

Recall: OLS chooses $\hat{\beta}_0$ and $\hat{\beta}_1$ to minimize the sum of squared residuals:

$$\min_{\hat{\beta}_0, \hat{\beta}_1} \sum_{i=1}^{n} \hat{e}_i^2 \quad = \quad \sum_{i=1}^{n} \left(Y_i \; - \; \hat{\beta}_0 \; - \; \hat{\beta}_1 X_i\right)^2$$

This gives us the **fitted values**:

$$\widehat{\text{wage}}_i \quad = \quad \hat{\beta}_0 \; + \; \hat{\beta}_1 \; \cdot \; \text{female}_i$$

β_1 is the truth; $\hat{\beta}_1$ is our guess from data

The population (all U.S. workers)

$\text{wage}_i = \beta_0 + \beta_1 \cdot \text{female}_i + \epsilon_i$

- β_1 = true average wage gap
- This is what we **want to know**
- Not a causal claim — just a difference

Our sample (526 workers)

$\widehat{\text{wage}}_i = \hat{\beta}_0 + \hat{\beta}_1 \cdot \text{female}_i$

- $\hat{\beta}_1$ = our best guess at β_1
- From *this particular* sample
- Different sample → different $\hat{\beta}_1$

Hats mean "estimated from this sample" — not the true population value

Because $\hat{\beta}_1$ comes from a sample, it could have been different

- Draw a different 526 workers from the CPS $\rightarrow$ get a different $\hat{\beta}_1$
- Maybe $-\$2.30$, maybe $-\$2.75$ — but probably close to the true β_1
- You will **never see** those other samples — you only have this one

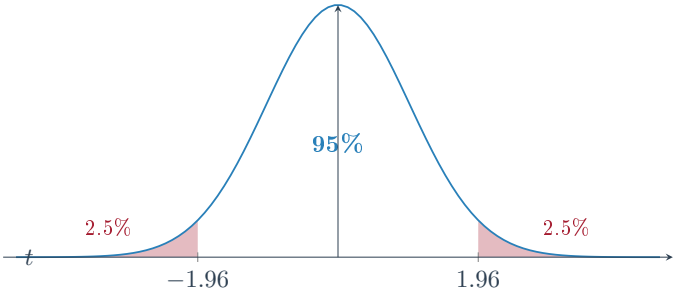
So how do you know where your $\hat{\beta}_1$
sits in that invisible distribution?

Your one sample contains enough information to estimate its own uncertainty

- **Standard error** $= \mathrm{SE}(\hat{\beta}_1)$
  Estimated *from your data* — tells you how wide the sampling distribution is
- **t-statistic** $= \hat{\beta}_1 \,/\, \mathrm{SE}(\hat{\beta}_1)$
  A thought experiment: "If the true gap were zero, how many SEs away is my estimate?"

$$t \;=\; \frac{\hat{\beta}_1}{\mathrm{SE}(\hat{\beta}_1)} \qquad — \qquad \text{large } |t| \Rightarrow \text{zero is implausible}$$

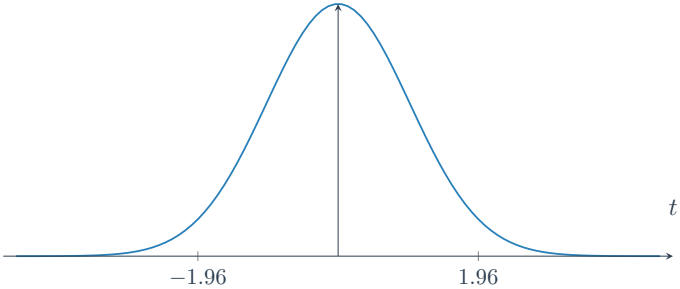
The t-distribution tells you which values are plausible under H_0



- Blue region: 95% of t-statistics fall between ± 1.96 if $\beta_1 = 0$
- Red tails: only 5% land here — these are "unlikely under H_0"

Our $t = -8.28$ is not even on the chart

Our t



$t = -8.28$ is so far from zero that the probability of seeing it under $H_0 : \beta_1 = 0$ is essentially zero

The p-value $= P(|t| \geq 8.28 \mid \beta_1 = 0) \approx 0.000000000000001$

With a binary X, what do $\hat{\beta}_0$ and $\hat{\beta}_1$ equal?

$$\widehat{\text{wage}}_i = \hat{\beta}_0 + \hat{\beta}_1 \cdot \text{female}_i$$

If female $= 0$ *(men), what is* $\hat{Y}$*?*

If female $= 1$ *(women), what is* $\hat{Y}$*?*

So what is $\hat{\beta}_1$*?*

$\hat{\beta}_0$ is the male mean; $\hat{\beta}_1$ is the gap

- Men (female = 0): $\hat{Y} = \hat{\beta}_0 \quad \Rightarrow \quad \hat{\beta}_0 = \overline{\text{wage}}_{\text{men}}$
- Women (female = 1): $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 \quad \Rightarrow \quad \hat{\beta}_0 + \hat{\beta}_1 = \overline{\text{wage}}_{\text{women}}$
- Subtract: $\hat{\beta}_1 = \overline{\text{wage}}_{\text{women}} - \overline{\text{wage}}_{\text{men}}$

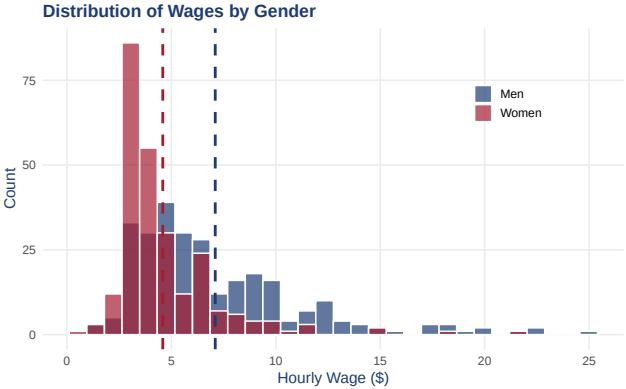
With binary X: intercept = baseline mean, slope = difference in means

Women earn \$2.51 less per hour than men on average

	Model 1
Intercept	7.10
	(0.21)
Female	**−2.51**
	(0.30)
R^2	0.116
n	526

- $\hat{Y}_{\text{men}} = 7.10 \quad \hat{Y}_{\text{women}} = 7.10 - 2.51 = 4.59$
- SE $= 0.30$ — if we resampled, $\hat{\beta}_1$ would typically move by $\pm\$0.30$
- $t = -2.51 \,/\, 0.30 = -8.28$ — our estimate is 8 SEs from zero

Is knowing someone's gender enough to predict their wage?



Look at the overlap. How much wage variation does gender explain?

$R^2 = 0.116$ means gender alone explains 12% of wage variation

- 88% of variation is *within* gender groups
- Some men earn very little; some women earn a lot
- Gender matters, but it is far from the whole story

$$R^2 = 1 - \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\sum \hat{e}_i^2}{\sum (Y_i - \bar{Y})^2} = 0.116$$

Turn to your neighbor

"The wage gap is $2.51."

What does that number **not** tell you?

Model 2: Adding Covariates

Model 2 adds education, experience, and tenure

$$\widehat{\text{wage}}_i = \hat{\beta}_0 + \hat{\beta}_1 \text{female}_i + \hat{\beta}_2 \text{educ}_i + \hat{\beta}_3 \text{exper}_i + \hat{\beta}_4 \text{tenure}_i$$

- Same outcome, same treatment variable
- Now we **hold constant** education, experience, and tenure
- This asks a *different question* than Model 1

Before we look: will the female coefficient get bigger or smaller?

Model 1: $\hat{\beta}_{\text{female}} = -2.51$

We are about to add education, experience, and tenure.

	Men	**Women**
Mean education	12.8 yrs	12.3 yrs
Mean experience	17.6 yrs	16.4 yrs
Mean tenure	6.5 yrs	3.6 yrs

Will the magnitude of $\hat{\beta}_{\text{female}}$ increase or decrease? Why?

The gender gap shrinks from \$2.51 to \$1.81 after adding controls

	Model 1	Model 2
Intercept	7.10	−1.57
	(0.21)	(0.72)
Female	−2.51	**−1.81**
	(0.30)	(0.26)
Education		0.57
		(0.05)
Experience		0.03
		(0.01)
Tenure		0.14
		(0.02)
R^2	0.116	**0.364**
n	526	526

Every coefficient has a "holding constant" interpretation

- **Female:** $-\$1.81$/hr, holding education, experience, and tenure constant
- **Education:** $+\$0.57$/hr per additional year, holding gender, experience, and tenure constant
- **Experience:** $+\$0.03$/hr per additional year, holding gender, education, and tenure constant
- **Tenure:** $+\$0.14$/hr per additional year, holding gender, education, and experience constant

"Holding constant" changes the question we are asking

Model 1

$\hat{\beta}_{\text{female}} = -2.51$

Gap between the *average* man and the *average* woman

Model 2

$\hat{\beta}_{\text{female}} = -1.81$

Gap between a man and woman with the *same* education, experience, and tenure

Adding covariates changes what $\hat{\beta}_1$ means

R^2 tripled — the model explains 3× more variation

- Model 1: $R^2 = 0.116$
- Model 2: $R^2 = 0.364$
- Education, experience, and tenure explain a lot of wage variation
- The SE on Female *shrank*: $0.30 \rightarrow 0.26$
  - More precise estimate when the model fits better

Turn to your neighbor: plug-in prediction

Using Model 2:

$$\hat{Y} = -1.57 + (-1.81)\text{female} + 0.57 \cdot \text{educ} + 0.03 \cdot \text{exper} + 0.14 \cdot \text{tenure}$$

Predict hourly wage for:

Profile: educ = 16, exper = 10, tenure = 5

1. A woman with this profile: $\hat{Y} = ?$

2. A man with this profile: $\hat{Y} = ?$

3. What is the gap?

Plug-in prediction: the gap is always $\hat{\beta}_{\textbf{female}}$

- **Man** (female $= 0$):
  $\hat{Y} = -1.57 + 0(-1.81) + 0.57(16) + 0.03(10) + 0.14(5) = \$8.54$
- **Woman** (female $= 1$):
  $\hat{Y} = -1.57 + 1(-1.81) + 0.57(16) + 0.03(10) + 0.14(5) = \$6.72$
- **Gap**: $6.72 - 8.54 = -\$1.81 = \hat{\beta}_{\text{female}}$

With no interaction, the gap is the same for every profile

Model 3: Interactions

Does marriage affect everyone's wages the same way?

Think about this:

- In 1976, who was more likely to be the primary earner in a married household?
- If employers perceive married men as "more stable," what happens to their wages?
- If employers perceive married women as "likely to leave," what happens?

*How would you **model** this — one slope for marriage, or two?*

Model 3 asks: does marriage affect wages differently for men and women?

$$\widehat{\text{wage}}_i = \hat{\beta}_0 + \hat{\beta}_1 \text{female}_i + \hat{\beta}_2 \text{married}_i + \hat{\beta}_3 (\text{female} \times \text{married})_i + \cdots$$

- $\hat{\beta}_2$ = effect of marriage **for men** (when female = 0)
- $\hat{\beta}_2 + \hat{\beta}_3$ = effect of marriage **for women**
- $\hat{\beta}_3$ = *difference* in marriage effect across genders

Marriage boosts men's wages by \$1.82 but *reduces* women's

	Model 1	Model 2	Model 3
Intercept	7.10	−1.57	−2.39
	(0.21)	(0.72)	(0.73)
Female	−2.51	−1.81	−0.31
	(0.30)	(0.26)	(0.41)
Married			1.82
			(0.40)
Female × Married			**−2.40**
			(0.53)
Education		0.57	0.55
		(0.05)	(0.05)
Experience		0.03	0.02
		(0.01)	(0.01)
Tenure		0.14	0.13
		(0.02)	(0.02)
R^2	0.116	0.364	**0.392**
n	526	526	526

The marriage premium: \$1.82 for men, −\$0.57 for women

Men (female = 0)

Marriage effect = $\hat{\beta}_2$
= +\$1.82/hr

Married men earn significantly more

Women (female = 1)

Marriage effect = $\hat{\beta}_2 + \hat{\beta}_3$
= $1.82 + (-2.40) = -\$0.57$/hr

Married women earn slightly less

$\hat{\beta}_3 = -2.40$ captures the *difference* in marriage premiums

Turn to your neighbor: 2×2 predictions

Using Model 3 with educ = 12, exper = 10, tenure = 5:

$$\hat{Y} = -2.39 + (-0.31)\text{female} + 1.82 \cdot \text{married} + (-2.40)\text{female} \times \text{married}$$
$$+0.55 \cdot \text{educ} + 0.02 \cdot \text{exper} + 0.13 \cdot \text{tenure}$$

Fill in the table:

	Single	**Married**
Men	?	?
Women	?	?

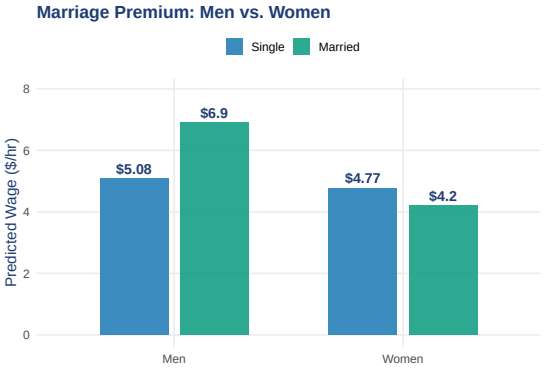
Which group earns the most? The least?

Married men earn the most; married women earn the least

	Single	Married
Men	$5.08	$6.90
Women	$4.77	$4.20

Marriage premium:

Men: +$1.82 Women: −$0.57



Putting It All Together

Track $\hat{\beta}_{\text{female}}$ across all three models

	Model 1	Model 2	Model 3
$\hat{\beta}_{\text{female}}$	-2.51 (0.30)	-1.81 (0.26)	-0.31 (0.41)
Interpretation	Raw gap	Adjusted gap (same human capital)	Gap for single workers (same HC)
R^2	0.116	0.364	0.392

The "wage gap" depends on which question you ask

Four skills to master before the exam

1. **Interpret coefficients in context**
   "One more year of education $\rightarrow$ +\$0.57/hr, holding ... constant"
2. **Compute $\hat{Y}$ by plug-in**
   Substitute values into $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \cdots$
3. **Compare models side-by-side**
   Track how coefficients, R^2, and SEs change
4. **Read interactions**
   $\hat{\beta}_3$ = difference in slopes $\rightarrow$ build the 2×2 table

Exam 1 is Thursday, March 12

- **Review guide:** posted on course website
- **Cheat sheet:** 2 pages, front and back, handwritten or typed
- **Practice:**
  - Plug-in predictions — get fast at arithmetic
  - Interpret every number in a regression table
  - Build 2×2 tables from interaction models

You have all the tools. Practice using them.